

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – SIH26002



Your
Team
Name

road-risk signals to an actionable logistics decision



❖ Proposed Solution

THE GAP: Disruption data is fragmented

- Flood / landslide risk
- Road accessibility
- Vehicle movement
- Field reports

OUR SOLUTION:

01 Sense - Rain + terrain + incidents

02 Predict - Road risk score

03 Route - Safer alternative

04 Act - Alert + field update

WHY IT STANDS OUT:

- NER-specific Terrain-aware logistics intelligence
- Explainable AI - Risk score with key contributing factors
- Offline-first - Field reports work during low connectivity

TECHNICAL APPROACH



- Technologies to be used
DATA SOURCES: Rainfall, Terrain/slope, Historical incidents, Field reports
INTELLIGENCE LAYER: Risk Engine (Random Forest + rule fallback), GIS + Routing (PostGIS + Dijkstra/A*), Offline Sync (PWA + IndexedDB queue), Alerts (Role-based notifications)
USER EXPERIENCE: Live risk map, Safer route, Vehicle view, Officer report

TECH			STACK:
•	React	+	Tailwind (Frontend)
•	FastAPI	+	Python (Backend)
•	PostgreSQL	+	PostGIS (Database)
•		scikit-learn	(ML)
•	Leaflet	/	OSM (GIS)
•	PWA	/	IndexedDB (Offline)
•	REST	APIs	(Integration)

FEASIBILITY AND VIABILITY



- Analysis of the feasibility of the idea
Build the core, validate the workflow, then scale

DATA: Open + simulated data - Prototype with realistic sample NER data; design APIs for future authorized feeds

AI: Explainable, not overbuilt - Random Forest risk model + rule fallback keeps the MVP fast and defensible

CONNECTIVITY: Offline-first reporting - Queue incident reports locally and sync automatically when the network returns

SCALE: Modular architecture - Add districts, users, data sources and alerts without redesigning the core

MVP FOR SIH:
• 3-4 sample NER districts

• Risk API
• GIS map

IMPACT AND BENEFITS



- Potential impact on the target audience
Turn fragmented information into faster logistics decisions

Benefits of the solution (social, economic, environmental, etc.)
ONE PLATFORM: Predict risk • Protect routes • Move essentials • Keep field teams connected

DISTRICT / DISASTER TEAMS: See high-risk road segments and field updates
LOGISTICS OPERATORS: Choose safer routes and monitor essential-goods movement
FIELD OFFICERS: Report incidents offline with GPS + evidence
PUBLIC / PARTNERS: Receive targeted, role-based disruption information

DEMO	SUCCESS	METRICS:
• Risk	response	< 2 sec
• Route	recompute	< 2 sec

- Offline sync
- Explainability

- Details / Links of the reference and research work
 - 01 MDoNER / NESIDS / PM-DevINE Programme context for North-East infrastructure and development
 - 02 AIKosh Open project data useful for prototype datasets and analytics
 - 03 Bhuvan / ISRO Geospatial context and terrain-related data sources
 - 04 Geological Survey of India Landslide and hazard context for risk modelling
 - 05 OpenStreetMap + SRTM Road network + elevation inputs for routing and terrain features

DATA INTEGRITY: Prototype $\neq$ live government system
Where granular live data is unavailable, the demo uses realistic simulated inputs.
The architecture remains ready for authorized integrations.

Design Principles:

✓ Explainable decisions
✓ Human approval for critical actions